

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

November 2013

## CL209 Structural Analysis-I

Date: 13.11.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

### Instructions:

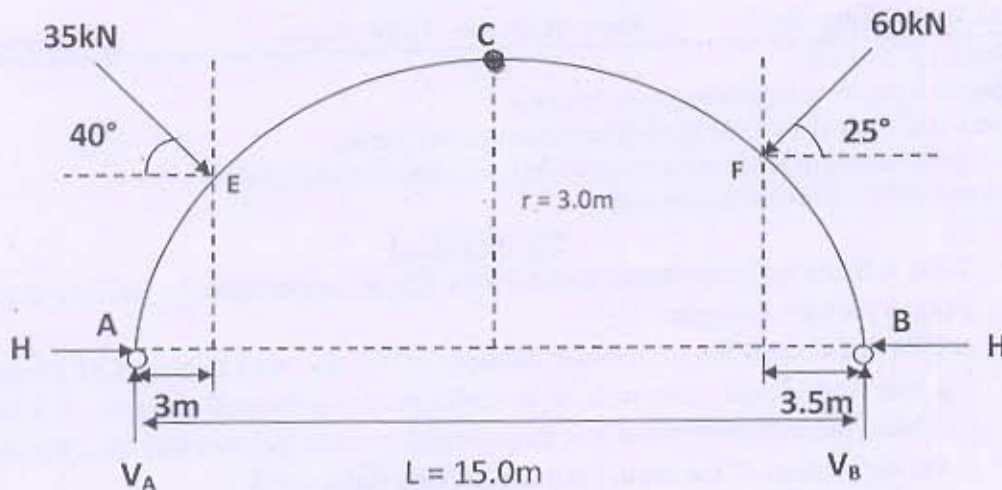
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

### SECTION – I

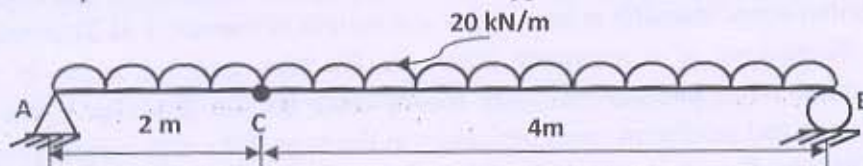
- Q – 1 What is Static Indeterminacy? Explain how you will determine the stability & SI for a plane truss with example. [07]
- Q - 2 (a) A cylindrical shell has an internal diameter of 1m and wall thickness of 15mm. It is 3m long and closed at its ends. It is subjected to an internal pressure of 3 N/mm<sup>2</sup>. Calculate the circumferential and longitudinal stresses induced and also the changes in the dimensions of the shell. Take  $E = 200\text{GPa}$  and  $\mu = 0.3$ . [08]
- (b) Attempt any One. [06]
- (i) A masonry chimney 18m high is of circular section, the external and internal diameter of the section being 6m and 3m respectively. The chimney is subjected to a horizontal wind pressure of 1500N/mm<sup>2</sup> of projected area. Find the maximum and minimum stress intensity at base. Take unit weight of masonry as 21kN/mm<sup>3</sup>.
- (ii) The dimensions of a masonry pillar are 8m high and 3m x 5m in section. A horizontal wind pressure of 1200 N/mm<sup>2</sup> acts on the face 3m x 8m. Find the maximum and minimum stress intensities at the base. Take unit weight of masonry is 20,000N/mm<sup>3</sup>.
- Q - 3 (a) A rectangular column of size 500mm x 250mm carries an eccentric load of 1000kN on the axis bisecting the thickness at 150mm from centroidal axis. Find maximum and minimum resultant stress and draw stress distribution diagram. [04]
- (b) Attempt any Two. [10]
- (i) Compare the strength of solid circular column of diameter 200mm and hollow circular column of same cross sectional area and thickness 30 mm. The other parameters are same for both of the sections.
- (ii) Determine the ratio of buckling strength of two columns, one hollow and other solid both are made of same materials and have the same length, cross sectional area, end conditions. The internal diameter of hollow column is  $\frac{2}{3}$ rd of external diameter.
- (iii) A column of circular section has 150 mm diameter and 3m length. Both ends of the column are fixed. The column carries a load of 100kN at an eccentricity of 15 mm from the geometrical axis of the column. Find the maximum compressive stress on the column section. Find also the maximum permissible eccentricity to avoid tension in the column section. Take  $E = 1 \times 10^5 \text{ N/mm}^2$ .

## SECTION - II

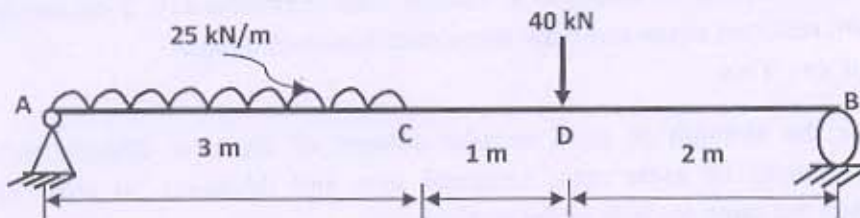
- Q - 4** A circular segmental three hinged arch has a span 12m and central rise of 3m is loaded as shown in Figure. Calculate reactions at the hinges and bending moments at the loaded points. [07]



- Q - 5** (a) A three hinged parabolic arch of 20m span and 4m central rise carries a point load of 6kN at 4m from left hand hinge. Calculate the support reactions. [04]  
 (b) Attempt any One. [10]  
 (i) Find out slope and deflection at  $x = 2$  m from support A. Take  $E = 6 \times 10^4$  kN.m<sup>2</sup>.



- (ii) By using Macaulay's method find slope at A and deflection at D of a beam as shown in Figure. Take  $E = 2 \times 10^5$  N/mm<sup>2</sup> and  $I = 3 \times 10^8$  mm<sup>4</sup>.



- Q - 6** Attempt any Two. [14]
- Two wheel loads of 16kN and 8kN at a fixed distance 2m apart, cross a beam of 10m span. Draw the influence line for B.M. and S.F. for a point 4m from the left abutment. Also find maximum B.M. and S.F. at that point.
  - A simply supported girder has a span 30m. Six wheel loads 50kN each, spaced 2m apart cross the girder from left to right. Find maximum BM at section 8m from the left hand support.
  - Find maximum SF and BM at section 4m from left end of simply supported beam of span 10m carrying UDL of 12kN/m on 3m length crossing beam from left to right.

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